A Topological Data Analysis of The 2024 Tacoma Equity Index

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Abstract

Topological Data Analysis (TDA) utilizes concepts from topology to analyze the shape of a dataset, and has been utilized for a diverse set of problems in fields, ranging from genomics and image analysis to economics and social science [2, 4, 6]. One of the tools we can use to perform TDA, is the *Mapper* Algorithm. For this exploratory analysis, we will be using this algorithm to interpret the 2024 Tacoma Equity Index, a high-dimensional data set that measures how various indicators shape equity outcomes for the city of Tacoma. With this methodology we produced results that showed inequity was generally more present in neighborhoods with populations higher in people of color. While this result was expected, it shows that TDA can be a legitimate avenue for analyzing equity in the city of Tacoma.

Keywords: Topological Data Analysis, K-mapper, Mapper algorithm, Tacoma Equity Index

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1 Introduction

The City of Tacoma's 2024 Equity Index dataset is a large, high-dimensional data set designed to inform equitable decision-making in policy, planning, and resource allocation by the City of Tacoma. It aggregates indicators across five broad equity categories: livability, accessibility, economy, education, and environmental health. This project performs an exploratory topological data analysis (TDA) of the Equity Index utilizing the *Mapper* algorithm from the Python library *K-Mapper*. This algorithm combines data-driven statistics with methods from algebraic topology to detect clusters, connectivity patterns, and outliers in the dataset. By applying these tools, our goal is to better understand how these factors shape equity outcomes for the city of Tacoma.

Traditional statistical models can summarize relationships among variables but often struggle in high dimensional datasets that have a lot of noise, missing data, and nonlinear relationships. TDA, in contrast, emphasizes the shape of data: it builds simplified network representations that preserve connectivity, clustering, and other structural features even in high-dimensional spaces. This approach has been successfully applied in fields ranging from genomics and image analysis to economics and social science. [2, 4, 6]

In this paper we first explain the Mapper algorithm to illustrate its functionality and familiarize the reader with ideas and vocabulary common in the growing field of topological data analysis. Then, we detail the process of collecting and cleaning the Tacoma Equity Index data using tools like JMP, R, and excel. Next, we use the K-Mapper Python library to walk the reader through the process of setting bounds for your analysis, understanding what a K-Mapper graph *means*, and analyzing some example graphs. Ultimately, this exploratory statistical analysis with topological methods aims to provide deeper insight into structurally complex datasets and support more equitable policy decisions.

2 Background

To examine the structure of a given dataset it is important to decide what sort of space to put the data in before we can examine it. A traditional embedding might have an n-dimensional dataset placed in $\mathbb{R}^n$. However, this can quickly run into issues off clarity to with the notion of connectedness and structure, specifically in datasets with a lot of noise. To avoid these issues we can choose a topological approach wherein structure and closeness are preserved even in noisy datasets. The differences between these embeddings for our use case can seem subtle, and in some ways they certainly are, but their effects on our ability to capture the structure of a dataset are huge. These advantages fuel our motivation for studying the 2024 Tacoma Equity Index via the lens of topological data analysis [1]. However, deeper understanding of these advantages are not necessary for understanding practical applications of topological data analysis and, thus, are beyond the scope of this paper. For further reading on the topic please see [3]. What is necessary, and will be explored in the following sections, is the Mapper algorithm that translate a dataset into a graph.

Before we begin explaining the mapper algorithm, we should first familiarize ourselves with the Tacoma Equity Index. We will do so in section 2.1 in which we describe the contents of the Tacoma equity index and explore some of the features of the dataset. This section is not meant to be a comprehensive overview of the Tacoma Equity Index but a preliminary exploration to

better understand the Tacoma Equity Index. We encourage readers to explore the dataset [8] and it's accompanying methodology report [5] themselves.

2.1 Exploratory Analysis

Before utilizing TDA, an initial exploratory analysis of the 2024 Tacoma Equity Index was conducted. The data set includes demographic data for race, age, and population for every census tract within the City of Tacoma. Along with these demographics, various measures for certain equity category were included. The methodology report states that the overall equity index value was calculated by aggregating five different equity categories:

1. **Livability Index**: Incorporates factors such as housing cost burden, general health, crime, and crashes, this category focuses on the quality-of-life aspects within the region.

2. **Accessibility Index**: This factors the ease of access to resources and services, highlighting the availability and proximity of essential facilities. Pavement condition and healthy food availability are two examples.

3. **Economy Index**: This focuses on financial factors such as jobs, poverty, and income, providing insights into the economic well-being of the census block.

4. **Education Index**: This evaluates the educational landscape within the region and includes educational attainment, student mobility, and testing proficiency.

5. **Environmental Health Index**: This category assesses the impact of the environment on the community's well-being. Such factors include diesel emissions and proximity to high traffic roads for example.

Even before our primary analysis, certain relationships between index values and other columns in the dataset. For example, here is a bubble plot that shows how the hispanic/latino demographic relates to the environmental index value:

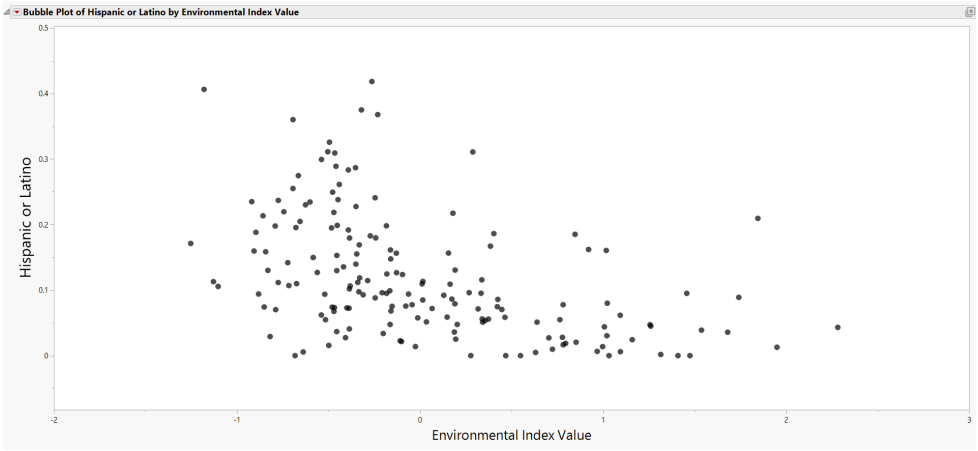

Figure 1: Bubble Plot

As we see in figure 1, there does seem to be some connection between the two columns, as it seems high environmental health index outcomes are less prevalent in census blocks with higher

latino demographic concentration. While this result is tangible, this is only visually displaying the relationships between the two columns specifically. Here is a scatterplot matrix involving multiple columns:

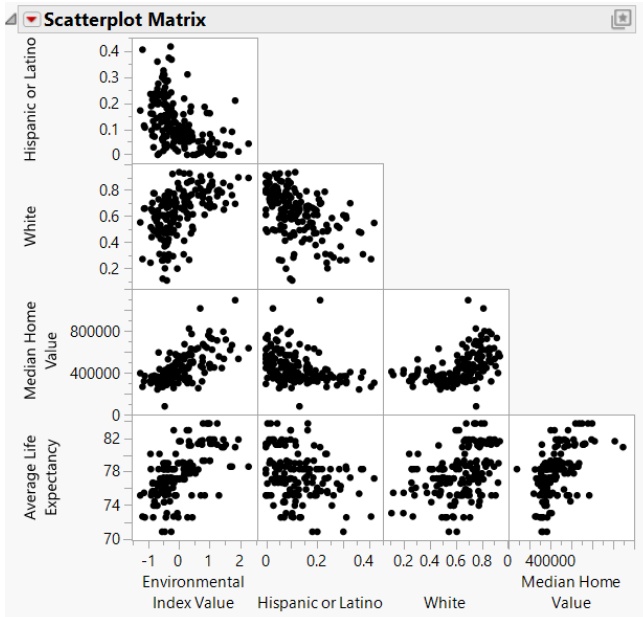

Figure 2: ScatterPlot

While visualizations like this can bring tangibility to higher dimensions of data, lets compare these methods to the Mapper algorithm in the following section.

2.2 Mapper Algorithm

With the mapper algorithm, there are four key steps to generating the graphs that we see in our results:

1. Lens Function Choice
   To start, we would like to project our high-dimensional data onto a lower dimensional space. We will do so by choosing a function which we will call our lens. Often times, our lens will be projected onto $\mathbb{R}$ or $\mathbb{R}^2$. In figure 3, we are shown a sample data set in $\mathbb{R}^2$ which is above the x-axis. Applying the lens function $f(x, y) = x$ projects the dataset onto the x-axis, reducing the dimensionality of the dataset.

Figure 3: An example dataset which we will project onto the x-axis below.

2. Cover Construction.
   After the dataset is projected into a lower dimension, our next goal is to create what is called a cover for the projected data. To do so, we choose a number of open sets (referred to as intervals in K-Mapper) and a percentage of overlap for adjacent open sets. We then lay these open sets over our our axis of projection such that all data falls within at least one open set and the open sets overlap each other by the percentage specified. This collection of open sets is called the cover and will be used later in the algorithm. Pictured in figure 4 are five open sets, each a different color, with a 33% overlap. This cover is defined for the axis of projection however they are pictured here extending into $\mathbb{R}^2$ to more accurately show which data falls in which open sets.

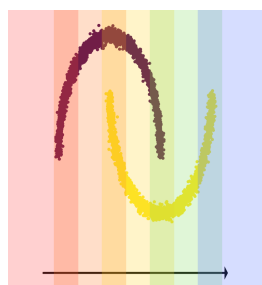

Figure 4: Cover chosen on the pre-image of the lens function

3. Clustering
   Next, we create clusters for all data in each intervals. To do so we implement a clustering algorithm called DBSCAN. Before we can run DBSCAN, however, we must choose a minimum cluster size and a radius (called an eps in K-Mapper). Once these parameters are set, DBSCAN begins classifying each point in the data. To do so it scans the radius around each point to determine which other points are nearby. If a point has more points within its radius than the minimum cluster size, the point is considered a core point. If a point has a core point within its radius but fewer points than the minimum cluster size within its radius, it is considered an edge point. Lastly, if a point does not have a core point within its radius and fewer points than the minimum cluster size it is considered noise and omitted from future clusters.

   Once each point has been identified, clusters are created for each core point and all edge points within the radius of each core point are added to their clusters. Then, the clusters

for core points within the radius of each other are combined so that all nearby core points are part of the same cluster.

4. Graph Generation
We now have a set of clusters for each interval and we would like to create a graph. To do so, we will created a node for each cluster and connect nodes by edges if their corresponding clusters share at least one common edge point. We are then left with a number of sub-graphs equal to the number of intervals. Then, connections are drawn between nodes in the sub-graphs if the nodes share any data points in the corresponding clusters. This has the effect of connecting our subgraphs at nodes in the overlap of our intervals. The final result is a graph where vertices represent clusters and edges represent shared data points between clusters, revealing the connectivity and "shape" of the data. The resulting graph for the dataset in figures one and two is pictured below. It is worth noting here that the graph generated can bare resemblance to the original data.

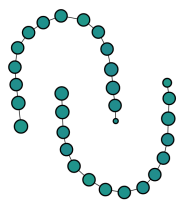

Figure 5: Graph generation using kmapper python library

3 Methods

Before results from the Mapper algorithm were gathered, there was a significant amount of data preparation, cleaning, and processing that had to be done to the Equity Index dataset. After this was completed, we then had to consider variations in specific KMapper parameters, that affected our overall output from the Mapper algorithm. This was done to detect stability issues with the generated graphs. Once this process was completed, we were able to finally yield results from the algorithm, the details of this process are detailed in the following subsections.

3.1 Adding Language Demographics

According to the methodology report, language demographics were collected from census data for 2022. We have implemented a step by step process that we have outlined in further detail in the appendix. First, we collected data based on the collection of GEOIDs within our original data set. After recording, we then calculated the z-scores for every specific language demographic. Since we have some overlap in the census tracts, we have the same z-scores for many of the GEOIDs. These empty spots were then sorted by the first three numbers of the GEOID which corresponded to the specific census tract that encompasses it. The interpretations were then performed in the normal fashion. This approach was deemed as a necessary step in order to analyse the effect that language had on the equity index.

3.2 Scaling the Data

For the issue of scale, our goal was to make it such that the distance between any two similar things in one column was similar to the distance between any two things in any other column. For example, the absolute difference between the maximum and median salary is considerably larger than the absolute difference between the max and median poverty rate. This was necessary so that the radius of our clustering algorithm could be fixed no matter the data we fed it. To achieve this, we chose to use a common statistical technique called z-score standardization. This form of standardization would make it so that each column has an average of zero and a standard deviation of one. Doing so allowed us to use radii near one in our clustering algorithm to consistently cluster related data points. This also encouraged easier analysis through means of more consistently generating graphs with lots of edges and nodes.

3.3 Fixing Polarity

Once our data was standardized, we changed the signs of certain columns to reflect that higher values of these columns were actually considered bad and lower values were considered good. For this, we once again referenced the Tacoma Equity Index methodology report which labels these kinds of columns as have a polarity such that "Lower is better" at the top of each column's description. These columns were for things like Diesel emissions and Ozone concentration where a higher amount of these things would indicate worse environmental factors. In total seventeen of these columns were identified and changed. It is worth noting as well, that a sign change was sufficient to reverse the polarity of these objects since we had already standardized.

3.4 Dealing with Missing Data

Finally, we had to choose how to deal with data points which were missing values. For one column, "Pedestrian Bicyclist Crashes" we chose to replace all missing data with zeros to indicate that no pedestrian bicyclist crashes had been reported in that specific census block at which point we re-standardized the data. This left us with far fewer rows that were missing data, a majority of which were missing four columns related to education data in census blocks near the borders of Tacoma. Weighing our options we decided the best course of action was selective listwise deletion of rows only when they were missing data from columns we were analyzing. For instance, a row missing data in the Average Student Mobility column will be omitted entirely from our data when we are using the Average Student Mobility column in our lens but the row will be included if we are not using the Average Student Mobility column in our lens.

We chose this route as opposed to replacing the data with the average or an estimate as is usually done for two main reasons. Firstly, since our data is z-score standardized replacing all empty cells with the average, which would in this case be zero, would lead to a considerable amount of artificial clumping around zero for all of these columns. Secondly, because of the low correlation these census blocks had with nearby census blocks, we decided that any linear regression techniques would be too inaccurate to use. There was consideration as well to remove the rows from analysis altogether, but since these rows were not higher in total population and had lower equity on average, their inclusion was necessary to have a full picture of the city of Tacoma as a whole.

3.5 Variations in Kmapper Parameters

After our data had been cleaned, we set bounds for variation in our Kmapper parameters. The specific parameters we set bounds for were the minimum size for a cluster, the number of intervals in the cover applied to our data and the size of their interval overlap, and the radius used in the clustering algorithm.

The number of intervals in our cover and the overlap of cover intervals varied between twelve and eighteen and 20% and 40% respectively. The number of intervals in our cover was chosen such that our standardized data, which largely falls within the range of -3 to 3, has two to four intervals per standard deviation of the selected lens. These parameters effect the number of nodes and edges and are varied in order to make the overall shape of the data more pronounced in the graph. Keeping these parameters within our predetermined range, however, allows us to ensure that all graphs we generate have the capability to be rendered in the same amount of detail.

The minimum number of samples in a cluster is always set to one when generating a graph. If the mininum number was set to anything other than one, there was a chance some census blocks would be left out of the final visualization. As such a minimum sample of one was done to ensure that the final Kmapper output had all census blocks contained somewhere in the visualization, even if it was just one single node.

The range of variation for the search radius (denoted eps) of the clustering algorithm is between 1 and $\sqrt{n}$ where n is the number of columns being fed into Kmapper. The goal of this search radius was to only have nodes be in the same cluster if they are within one standard deviation of each other. Recalling section 3.3, because of our standardization technique during data processing, the standard deviation of each column is 1. This means that the upper bound of this parameter must change in accordance with the dimensionality of the data we are giving to Kmapper. For example, an analysis involving 6 columns like that in Section 4.2 would necessitate a larger eps, where as the analysis conducted in Section 4.1 used only 4 columns, and as such used a lower eps setting. Within these bounds, once again, we vary this parameter to ensure the shape of the data is pronounced while maintaining a similar amount of detail once we've rendered the graph.

3.6 Note on Stability Testing

Lastly, after the bounds for variation were set, we developed a method to test the stability of our model. We did so by generating each graph 27 times, gradually adjusting the 3 parameters detailed in Section 3.5 slightly each time. This was done to ensure that minor changes in our parameters did not produce major changes in our output. To evaluate these changes we laid out criteria to quantify these changes. First we found the difference in the count of path connected components which had more than eight nodes. Next we found the difference in the count of total path connected components. Lastly, we found the difference in count of nodes attached to large path connected components but are not included in the longest possible non-overlapping path. A dramatic change in any of these counts could have indicated an issue with the stability of our model. Outputs with potential stability issues detected in this phase of generation were not chosen to be used as results.

4 Results and Discussion

In the following section we will produce kmapper results from the Tacoma Equity Index. With our newfound understanding of topological data analysis we will also interpret our results to find what the Tacoma Equity Index says about the Tacoma.

4.1 First Analyses of KMapper

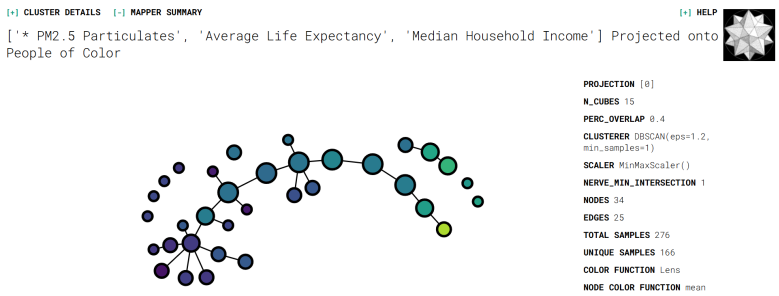

Figure 6: Various categories projected onto People of Color.

For our first example we will note three important features which are communicated to us somewhere in the visualization. First, we must know what census block statistics are being used in our visualization. The statistics being used here, are mentioned by name in the title of the visualization, in this case they are "PM2.5 Particulates" which is the concentration of particles less than 2.5 micrometers in the air, "Average Life Expectancy", "Median Household Income", and "People of Color" which is the proportion of people of color in a given census tract. Second, we will want to identify the lens function which is typically one of the statistics but can be any number of things. In this case, our lens is "People of Color" which is indicated in the title with the phrase "projected onto" prefacing it. Thirdly, we want to identify the way in which the nodes were colored. In the "MAPPER SUMMARY" side panel, the "COLOR FUNCTION" label identifies the lens as the coloring scheme of our nodes. Armed with this information we can begin to read the visualization and understand what it says about how our lens relates to our other statistics.

We will begin by examining our largest group of connected nodes which, due to its mostly linear structure, we will often refer to as a spine. At the right-hand side of the spine, we find lighter yellow, green, and blue nodes which represent higher percentages of people of color as per our color function. We notice that at this end of the spine, there are no nodes which deviate from the main spine. This means that the census block data we are examining tends to be similar for all of these lighter colored nodes. When opening the cluster details panel for all of these clusters, we see that this is true and that census blocks with higher populations of people of color tend to be below average in all other statistics. To be exact, the other statistics tend to be less than one standard deviation below the mean value for Tacoma.

The opposite end of the spine, on the other hand, has a comparatively large number of offshoots and has darker colored purple and blue nodes indicating a lower population of people of color (i.e. higher population of white people.) These offshoots lead us to conclude that there are a variety of census blocks that have a relatively low population of people of color whose

data for our other census block statistics form multiple clusters which are near to each other but distinct. Once again, we open our cluster details and find that many of these nodes furthest to the left feature clusters with above average values for most of, if not all, of these statistics but happen to have extremely high values for one or two statistics. For instance, we can review the difference between the dark purple node and the dark blue node which it is connected to. Both of these nodes have populations of people of color about one and a half standard deviations below the mean for Tacoma and PM 2.5 particulate and life expectancy values one standard deviation above the mean for Tacoma. These two nodes fail to cluster together, however, because the average median household income for the purple node is two standard deviations above the mean for Tacoma while the same statistic for the blue node is only one standard deviation above the mean for Tacoma. The offshoots towards the center of the spine are not dissimilar to the offshoots on the left-hand side of the spine, however their analysis will be left as an exercise for the reader.

Lastly, we will consider some of the node groups and individual nodes which lay outside the spine. Specifically, we will be examining the lighter colored nodes with high populations of people of color. Examining the group of three nodes, we find that this group is similar to the nodes on the right-hand side of the spine. They fail to connect to the spine, however since these nodes feature even lower values for all three other statistics than those on the spine. This is also the case for one of the two light-colored nodes which is disconnected from the spine. This conclusion about our disconnected nodes is verified during stability testing by noting that the census blocks in these nodes are able to cluster with other nodes under different parameters. The other disconnected light colored node, however, features a high population of people of color as well as above average values for all other statistics. As expected, this node stays disconnected under varying parameter.

Now that we have spent some time getting used to the methods of analysis for a graph generated by KMapper, we are prepared to examine other KMapper graphs. In the next two sections we will do exactly that with two more graphs generated from the data in the Tacoma Equity Index.

4.2 KMapper Analysis Using White Population as the Lens Function

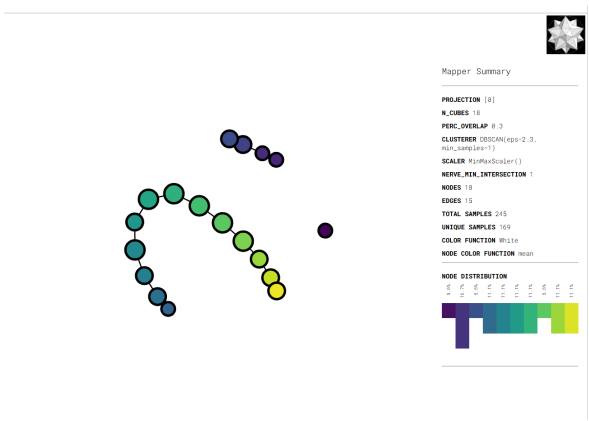

Figure 7: All Indices projected onto the White Demographic.

For this KMapper analysis, we set the lens function to correspond to the White demographic. The same color gradient from section 4.1 was used as the visual indicator. The other columns that were projected onto this lens were the 5 indices utilized to calculate the overall equity index column for the dataset: The Economic Index, Education Index, Livability Index, Accessibility Index, and Environmental Index. The resulting graph formed a long, connected spine that has 13 nodes, a smaller disconnected spine with only 4 nodes, and a single disconnected node. The 13 node spine showed a continuum in how the white demographic relates with the equity indices, where census blocks with higher white population tended to have higher equity outcomes. It should be noted that the equity index calculates the "people of color" column as any nonwhite demographic, so this K-Mapper visualization also shows that areas with higher non-white demographics have lower equity incomes. As such if "people of color" was chosen as a lens function, the *Mapper* algorithm would yield the exact same graph as seen in Figure 7 structurally. The only difference would be an inverted color scheme highlighting that equity is lower as non-white population increases.

On the other-hand, the disconnected smaller spine indicated a small cluster where White demographic proportions behave differently relative to the 13-node spine. What exemplifies this difference is that the clusters in this path component have white populations that are 1.5-2.5 standard deviations below the mean, but the main indices were higher on average. A specific block from this cluster, GEOID: 530539400084 from the 9400.08 census tract shows how this cluster is an outlier, as shown in 8a and 8b below:

(a) GeoID 530539400084 from the 4-node cluster, highlighted by a red outline

CLUSTER STATISTICS

ABOVE AVERAGE

Feature	Mean	STD
Education Index Value	0.656	1.1
Livability Index Value	0.647	1
Accessibility Index Value	0.355	1
Economic Index Value	0.469	0.9
Environmental Index Value	0.163	0.2

BELOW AVERAGE

Feature	Mean	STD
White	-1.729	1.7

(b) Cluster statistics on this isolated GEOID

From these figures, we see that while the white population in this block is lower than average, the indices are above average in contrast. The population of this node is also significant, as the total population for this census block is 3.14 standard deviations higher than the average. This is a stark contrast when comparing it to the census blocks clustered within the main spine, where there is a direct correlation between low-white population and low equity.

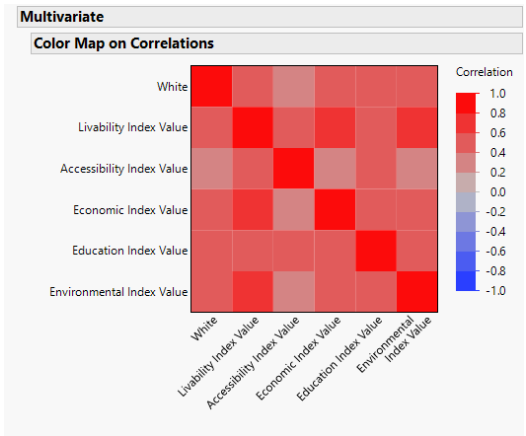

Figure 9: Correlation Matrix relating the Indices to the White demographic.

The results we see in the KMapper visualization are also supported by the values seen in the correlation matrix for the columns. We see that areas with higher proportions of White population tend to score higher in Environmental, Educational, and Economic indices. Accessibility is the least aligned, suggesting that White demographic concentration is not strongly tied to how accessible services or resources are. Livability has a modest but consistent correlation, meaning communities with higher White populations also tend to report higher livability outcomes. This relationship is visualized in the KMapper graph as one consistent 13 node spine, with lighter colors emphasizing how positively correlated white population proportion is with all indices. One thing that is not represented in this matrix however, is outlier points like that shwon in figure 6, that are clearly visible in the KMapper visualization.

Overall, the KMapper graph shows a connected path that represents how the White demographic proportion transitions smoothly across the Tacoma Equity indices. The correlation matrix reinforces that these transitions align most closely with environmental quality, education, and economic outcomes, while accessibility diverges by a small amount. The disconnected branch in the KMapper output corresponds to outlier regions where White population proportion does not align with typical relationships with the indices. This suggests the presence of neighborhoods where demographic-index relationships differ and could warrant deeper investigation. Taken together, this suggests that the White demographic serves as a strong and stable organizing axis in the dataset, and visually exemplifies how non-white demographics experience inequity.

4.3 Language Demographics

This version of the data set had some disconnected clusters which tells us they share no points across any of the partitions on the pre-image of the data. These disconnected clusters were retained in the final graph. The outputs of each of the k-mapper analyses were placed on a grid. The nodes were ordered from left to right in ascending order according to the English ZSCORE value. The clusters were then placed on the grid lines based on the number of "Above Average" indices each cluster contained. The table below shows the proportion of the population who speak each language.

Using a specific language as a lens gives a different viewpoint of the indices. Looking at the

data in table 1 directly, we can see there are many more English speakers (84.3%) than other languages, with Spanish as a distant second as 6.3%, as provided in 1 Since English speakers are by far the majority in Pierce county, we use this as the data set we would like to analyze. [1]

For stability testing-, we set various eps values between 2.2, 2.3 and 2.4 and n values between 12, 15 and 18. For each of the outputs, no significant deviations in connectivity were observed. The loss of detail followed a predictable pattern in which outliers were added to other nodes with no new spines or connectivity in any of the varied graphs.

For the lens of the Mapper algorithm, we chose the English Language ZSCORE. This collapses all of the data in the cloud onto the English language axis as a normalized score. The parameters we looked at were:

- Accessibility Index
- Economic Index
- Education Index
- Environmental Index
- Livability Index

Language	***Percentage***
English	84.3%
Spanish	6.3%
French, Haitian, or Cajun	0.4%
German or other West Germanic languages	0.4%
Russian, Polish, or other Slavic languages	1.4%
Other Indo-European languages	0.7%
Korean	1.3%
Chinese (incl. Mandarin, Cantonese)	0.4%
Vietnamese	0.7%
Tagalog (incl. Filipino)	1.2%
Other Asian and Pacific Island languages	2.1%
Arabic	0.1%
Other and unspecified languages	0.8%

Table 1: Proportion of languages spoken in Pierce County

[1] For brevity and content in this paper, simplifications were made to reduce the amount of content. We treat the compliment of the event "one speaks English" as "one does not speak English". In this context, it is an oversimplification on the basis that language is not based on a binary, but further analysis would render this section far too long.

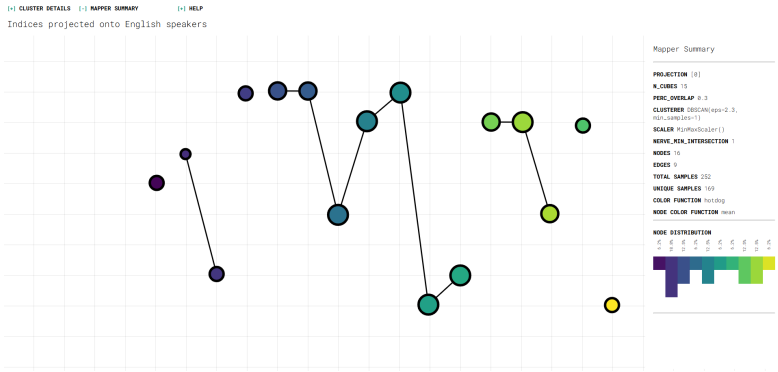

Figure 10: Kmapper output with English Speaker Lens

Let each of the major spines in the English graph be labeled from left to right, E_1, E_2 and E_3, respectively. Taking the average of the mean and the standard deviation of all nodes in each spine, gives us the following table:

	E_1		E_2		E_3	
	Mean	***STD***	***Mean***	***STD***	***Mean***	***STD***
English	-1.69	1.90	1.56	1.33	-0.37	0.73
Accessibility	-0.18	1.00	0.16	0.47	-0.03	0.20
Economic	-0.08	0.55	0.23	0.40	-0.03	0.27
Education	-0.44	0.70	0.21	1.00	-0.09	0.29
Enviromental	-1.18	1.65	0.47	0.73	0.00	0.40
Livability	-1.14	1.85	0.16	0.33	0.03	0.26

Table 2: Spine centrality data from Kmapper output

According to the correlation probability and matrix in figure 11, the English Z-score and Standardized equity have a 55% chance to be correlated at 0.45, or weakly positively correlated. This means that as the English-speaking z-score increases in an area, we can expect the overall equity index to increase as well. We also find with English speakers that there is generally a weak positive correlation with the indices.

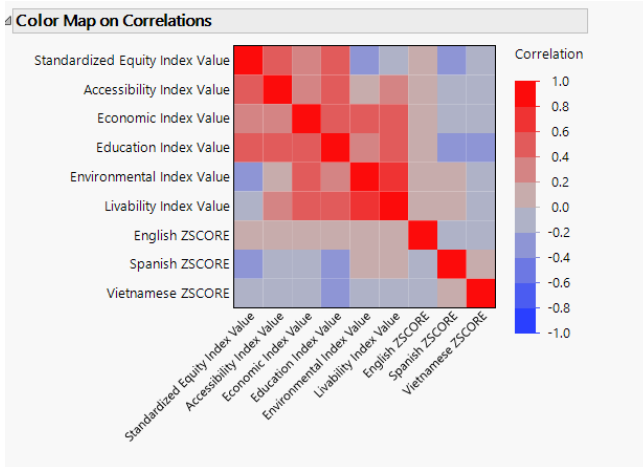

Figure 11: Heat map of correlations created in JMP, red is a positive correlation, blue is a negative correlation.

First, we note the descending order of English speakers in each of the spines, the order being E_2, E_3, E_1. Where the mean is almost 2 standard deviations between E_1 and E_2. Comparing the rest of the indices listed, we see that all indices in E_1 are negative, and opposite for E_2. This could necessarily indicate that there are some resource disparities where E_1 could be lacking in education or accessibility resources and E_2 has ample services or resources available to them. To simplify the interpretation, if we assume that a low amount of English speakers implies there is a larger amount of non-English speaking people, we can then say that the less English speakers there are in an area, the lower the Livability and Environmental indices become.

We used these versions of software to complete this project.

Jupyter	v1.1.1
K-Mapper	v2.1.0
JMP	Pro 18

5 Conclusion

Using topological data analysis in combination with the Tacoma Equity index, we were able to group and organize the neighborhoods of Tacoma by demographic information, equity indices, and individual quality of life indicators. Once organized, we were able to explore our groupings to find common themes across similar neighborhoods as well as neighborhoods which seem to be doing unexpectedly good or bad considering which neighborhoods they were most similar too. This investigation into of the Tacoma Equity Index unilaterally revealed evidence of inequity across the neighborhoods of Tacoma. In particular, it was revealed that lower indices were generally biased towards neighborhoods with more people of color and with less english speakers. These findings are exactly what we would expect considering Tacoma's history of discriminatory practices [7] which still have wide reaching effects to this day.

It is important to note that the ability to corroborate our existing understanding of equity in Tacoma is twofold. For one, a new lens on old data, always has potential to reveal deeper

understanding. Our analysis could provide yet another avenue for addressing inequity in Tacoma. Additionally, it also serves to illustrate the usefulness of topological data analysis in a world of information. With so many problems to address and so much information, the biggest constraint on action is understanding. Topological data analysis provides that in new and unique ways. While the results of this exploratory analysis aren't surprising, the clarity of the results given by topological data analysis showcases its usefulness, and potential to be utilized in future problems involving large datasets.

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6 Appendix

6.1 Language Demographics

In the methodology report, a source was given for language spoken demographics, yet there was no data given in the original dataset. The source given was from a census in 2022, so there was collection and cleaning to do on this set of data.

6.1.1 Collect data from GEOID20 and census tract

The GEOID20 is a 15 digit number assigned to the census areas of the US, where there is information coded into it such as the geographical location and arbitrary location about census data. The last 5 numbers represent the census tract (the first three) and the specific partition where the GEOID20 differs from the census tract. The census tract was extracted from each of the assigned GEOID20 numbers and recorded.

6.1.2 Recording the Raw Data from Census

The data given in the census record was given in estimates with appropriate errors mentioned on the spreadsheet. The decision we made was to get rid of the estimate over/underage and just take the estimate as recorded in the column labeled "estimate".

6.1.3 Z-Score calculation

Using the excel formula "standardize", the raw data was standardized using the first instance of the census tract, for example for census tract 602 there are two different GEOID20s that end in $602xx$. We record the standardized Z-Score for the first instance and leave the second one blank. This is repeated for all of the languages and all of the initial instances of a census tract.

6.1.4 Fill in the Blanks

The subsequent instances of the same census tract will share the same standardization score, since the census tract does not share the same partitions as the GEOID20 number. Thus, for every blank space, we assign the same z-score to each by copying from above and pasting below so that every blank space has a value.

6.2 Missing Data Analysis

While deciding on what to do with missing data, we noted 16 census blocks that had 4 or more missing columns. The most common missing columns shared between the data were the following education indicators: Average Student Mobility, Average Testing Proficiency, High School Graduation Rate, and Kindergarten Readiness Rate. Some of these rows had missing values in other columns as well such as Average Pavement Condition, Healthy Food Availibility, and Tacoma Personal Crime. To further explore these rows with missing values, we looked up the census block associated with the row by the GEOID. What we found was that 8 of these rows included low-equity census blocks that were not incorporated into the city of Tacoma, and areas with close proximity to highways. In particular, census blocks near Fife, Midland, and

Parkland made up these 8 rows. There other 8 rows were exceptions to this, as there were high-equity census blocks around University Place and Titlow Park that also had consistent missing columns in the education index indicators shown in figure 4.

The unincorporated census blocks in the south and east of the city on average have very low equity compared to the rest of Tacoma. As shown in the table, this discrepancy is not due to low population, since the row with GEOID 530530716012 has a total population of 1,481, which is 0.59 standard deviations higher than the mean population for all census blocks. For all other census blocks in these outliers, the population were around 1 standard deviation away from the mean with exception to the two census blocks making up Fife, which had standard deviations of 2.4 and 1.5 lower than the mean for the south and north respectively, and the census block 530530716014 which was 1.66 standard deviations lower than the mean. Since most of these rows were 1 or less deviations from the mean total population, we found that their inclusion was necessary and implemented the list-wise deletion method mentioned in section 3.5.

Table 3: Census Blocks with missing columns.

CBlock Group	Avg Pavement	Healthy Food	Avg Stud Mob	Avg Test Prof	HS Grad Rate	Kinder Ready	Tacoma Per Crime
530530716012	0.071	1.393	•	•	•	•	−0.127
530530716014	0.815	0.925	•	•	•	•	1.153
530530717031	0.627	−0.620	•	•	•	•	−0.689
530530717032	−2.234	−1.393	•	•	•	•	−0.532
530530717051	1.732	0.152	•	•	•	•	−0.734
530530723054	−2.142	0.410	•	•	•	•	−0.779
530530723092	1.639	1.853	•	•	•	•	−0.667
530530723094	−0.759	−0.041	•	•	•	•	−0.756
530530723101	−2.142	0.410	•	•	•	•	−0.689
530530723111	•	0.152	•	•	•	•	•
530539400021	0.256	4.016	•	•	•	•	−0.689
530539400022	2.653	0.410	•	•	•	•	−0.622
530539400023	−0.113	0.271	•	•	•	•	−0.779
530539400024	0.356	−0.041	•	•	•	•	•
530539400044	3.378	•	•	•	•	•	−0.734
530530723093	1.362	−1.393	•	•	•	•	−0.734

Table 4: The 16 rows with their standardized total population.

Census Block Group	**Equity Index Value**	**Total Population**
530530716012	−0.264	0.586
530530716014	−0.448	−1.662
530530717031	−0.869	2.046
530530717032	−0.539	0.761
530530717051	−0.660	−0.944
530530723054	0.239	−0.842
530530723092	0.571	−0.731
530530723094	0.361	−0.602
530530723101	0.371	−0.440
530530723111	0.001	1.413
530539400021	0.135	−0.177
530539400022	−0.053	−2.403
530539400023	−0.327	−1.513
530539400024	−0.028	0.683
530539400044	0.026	−0.332
530530723093	0.541	0.119